

AI/ML Internship – Day 61 Notes & Tasks

Module 8: NLP in Chatbots – Intent Recognition, Entity Recognition & Text Preprocessing

Part 1: Recap of Day 60

Yesterday we learned:

-  Menu-Driven Chatbot
-  College Enquiry Chatbot
-  Python Dictionary
-  User Input Handling

Today we will learn how **AI Chatbots understand human language** using **Natural Language Processing (NLP)**.

Part 2: What is NLP?

Definition

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, process, and generate human language.

It allows machines to understand what humans speak or type.

Simple Definition

NLP is the technology that helps computers understand human language just like humans do.

Real-Life Examples

- ChatGPT
- Google Assistant
- Siri
- Alexa
- Customer Support Chatbots

- Voice Assistants
 - Google Translate
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📌 Part 3: Why Do Chatbots Need NLP?

Imagine a customer types:

I forgot my password.

The chatbot should understand:

The customer wants to reset the password.

Without NLP:

✗ Chatbot sees only text.

With NLP:

✓ Chatbot understands the meaning.

📌 Part 4: Chatbot NLP Workflow

User Message

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Text Preprocessing

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Intent Recognition

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Entity Recognition

|



Response Selection

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Reply to User

📌 Part 5: What is Intent?

Definition

Intent is the **purpose or goal** behind a user's message.

Simple Meaning

Intent answers the question:

"What does the user want?"

Example 1

User:

I forgot my password.

Intent:

Password Reset

Example 2

User:

Track my order.

Intent:

Order Tracking

Example 3

User:

Book a train ticket.

Intent:

Ticket Booking

Part 6: More Intent Examples

User Message	Intent
Hello	Greeting

User Message	Intent
Bye	Goodbye
What is today's weather?	Weather
Book a hotel	Hotel Booking
Cancel my order	Order Cancellation
Show my balance	Balance Enquiry

Part 7: What is Entity?

Definition

An Entity is important information extracted from a user's message.

Simple Meaning

Intent tells **what the user wants**.

Entity tells **important details** about the request.

Example

User:

Book a flight to Delhi tomorrow.

Intent:

Flight Booking

Entities:

Destination → Delhi

Date → Tomorrow

Another Example

User:

Transfer ₹5000 to Rahul.

Intent:

Money Transfer

Entities:

Amount → ₹5000

Receiver → Rahul

Part 8: Intent vs Entity

Intent	Entity
User's goal	Important information
One per message (usually)	Can be multiple
Book Ticket	Delhi, Tomorrow
Order Tracking	Order ID
Hotel Booking	City, Date

Part 9: Text Preprocessing

Before understanding a sentence, chatbots clean the text.

This process is called **Text Preprocessing**.

Workflow

Raw Sentence



Lowercase



Tokenization



Stop Word Removal



Stemming/Lemmatization



Intent Detection

Part 10: Lowercasing

Example:

HELLO



hello

Why?

So that:

Hello

HELLO

hello

are treated as the same word.

Part 11: Tokenization

Definition

Tokenization means splitting a sentence into individual words (tokens).

Example:

Sentence:

I love AI.

After Tokenization:

```
["I", "love", "AI"]
```

Python Example

```
text = "I love AI"
```

```
tokens = text.split()
```

```
print(tokens)
```

Output:

```
['I', 'love', 'AI']
```

Part 12: Stop Word Removal

Some words appear frequently but carry little meaning.

Examples:

is

the

am

are

of

to

and

These are called **Stop Words**.

Example:

Sentence:

I want to book a ticket.

After removing stop words:

want

book

ticket

Now the chatbot focuses on the important words.

Part 13: Stemming

Definition

Stemming reduces words to their root form by removing prefixes or suffixes.

Examples

Original Stem

Playing Play

Played Play

Playing Play

Running Run

Python Example

```
from nltk.stem import PorterStemmer
```

```
stemmer = PorterStemmer()
```

```
print(stemmer.stem("playing"))
```

Output:

play

Part 14: Lemmatization

Definition

Lemmatization converts a word into its correct dictionary (base) form.

Examples

Word	Lemma
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Better	Good
--------	------

Running	Run
---------	-----

Children	Child
----------	-------

Mice	Mouse
------	-------

Unlike stemming, lemmatization produces meaningful words.

Part 15: Stemming vs Lemmatization

Stemming	Lemmatization
Faster	Slower
Removes suffixes	Uses dictionary
May produce invalid words	Produces valid words
Less accurate	More accurate

Part 16: Complete Chatbot Example

User types:

Book a train ticket to Mumbai tomorrow.

Step 1

Lowercase

book a train ticket to mumbai tomorrow

Step 2

Tokenization

book

a

train

ticket

to

mumbai

tomorrow

Step 3

Remove Stop Words

book

train

ticket

mumbai

tomorrow

Step 4

Intent

Book Ticket

Step 5

Entities

Destination → Mumbai

Date → Tomorrow

Step 6

Response

Your train ticket booking has started.

📌 Part 17: Python Example

```
text = "Book a train ticket to Mumbai tomorrow"
```

```
text = text.lower()
```

```
tokens = text.split()
```

```
print(tokens)
```

Output

```
['book', 'a', 'train', 'ticket', 'to', 'mumbai', 'tomorrow']
```

📌 Part 18: Real-World Applications

Banking Chatbots

Balance enquiry

Money transfer

Loan enquiry

Healthcare

Book appointment

Medicine reminders

Doctor consultation

E-Commerce

Track order

Return product

Product search

Education

Student support

Attendance

Course details

Travel

Flight booking

Hotel booking

Cab booking

Part 19: Advantages of NLP

-  Understands user language
 -  Identifies user intent
 -  Extracts important information
 -  Improves chatbot accuracy
 -  Enables natural conversations
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Part 20: Limitations

-  Difficult with spelling mistakes
-  Different languages require different models
-  Complex sentences may confuse the chatbot
-  Requires training data for AI-based systems

Part 21: Theory Questions

What is NLP?

NLP is a branch of AI that enables computers to understand and process human language.

What is Intent?

Intent is the goal or purpose behind the user's message.

What is Entity?

An entity is important information extracted from the user's message.

What is Tokenization?

Splitting a sentence into individual words.

Difference between Stemming and Lemmatization?

Stemming removes suffixes, while lemmatization converts words to their correct dictionary form.

Part 22: Interview Questions

Why is NLP important in chatbots?

NLP helps chatbots understand user language and respond intelligently.

What is the difference between Intent and Entity?

Intent identifies what the user wants, while entities provide important details related to that intent.

Explain Tokenization.

Tokenization is the process of breaking a sentence into smaller units called tokens.

Why remove Stop Words?

Stop words are removed because they do not contribute much meaning and can reduce processing efficiency.

Give examples of NLP applications.

- Chatbots
 - Voice Assistants
 - Google Translate
 - Spam Detection
 - Sentiment Analysis
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Tasks for Day 61

Theory

1. Define NLP.
 2. Explain Intent Recognition.
 3. Explain Entity Recognition.
 4. Differentiate Stemming and Lemmatization.
 5. Explain the NLP workflow in chatbots.
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Practical Task 1

Convert the following sentence to lowercase:

HELLO, I WANT TO BOOK A TICKET.

Practical Task 2

Tokenize the sentence:

I love learning Artificial Intelligence.

Practical Task 3

Identify the intent and entities:

Book a flight to Chennai on Friday.

Practical Task 4

Write a Python program to tokenize a sentence using `split()`.

Practical Task 5

Create a table showing the difference between Stemming and Lemmatization.